COVER LETTER FOR SUBMISSION

Date: 6 November 2026

To:
The Editor-in-Chief
Remote Sensing Applications: Society and Environment (RSASE)
Elsevier B.V.

Subject: Submission of Original Research Article titled *"The Governance Void at Concession Frontiers: A Spatial Boundary Discontinuity Analysis of Peatland Wildfires in Central Kalimantan"*

Dear Editor-in-Chief and Editorial Board Members,

We are pleased to submit our original research manuscript entitled **"The Governance Void at Concession Frontiers: A Spatial Boundary Discontinuity Analysis of Peatland Wildfires in Central Kalimantan"** for consideration for publication as a regular research article in *Remote Sensing Applications: Society and Environment* (RSASE).

Journal Fit and Core Contribution to Remote Sensing, Society, and Environment

Tropical peatland wildfires across equatorial Southeast Asia present one of the most pressing interdisciplinary environmental crises in the world, directly bridging physical remote sensing technologies, ecosystem degradation, and societal land governance. While statutory environmental enforcement in Indonesia emphasizes corporate strict liability (Law No. 32/2009, Article 88), recurrent megafires continue to release catastrophic carbon emissions and regional particulate matter (PM2.5) smoke-haze during drought pulses.

Remote Sensing Applications: Society and Environment provides the ideal home for this research because our study operationalizes satellite Earth observation data to directly interrogate the societal, tenurial, and ecological dynamics governing tropical landscapes: 1. **Satellite Remote Sensing Integration:** We integrate 43,937 verified satellite thermal anomalies from the NASA FIRMS constellation (VIIRS I-Band 375-m and MODIS 1-km active fire products), Fire Radiative Power (FRP), 13 primary canal vectors digitized in metric UTM Zone 49S projection, and ECMWF ERA5-Land atmospheric reanalysis. 2. **Quasi-Experimental Spatial Boundary Discontinuity Design (Spatial BDD):** Rather than treating satellite fire observations as unstructured point clouds, we implement a Spatial BDD across $\pm 3,000$ meters of 812 audited cadastral concessions (627 oil palm HGU and 185 commercial forestry PBPH). We reveal that active fire density jumps sharply by **+56.5%** directly across certified borders onto unallocated state land (*Area Penggunaan Lain*, APL), exposing a severe institutional governance void driven by private boundary surveillance combined with unmanaged trans-boundary peat drainage desiccation. 3. **Tobler's First Law and Spatial Machine Learning:** We demonstrate that tree-based remote

sensing models ingesting geodetic coordinates suffer from spatial memorization and data leakage. Under 5-fold Spatial Block Cross-Validation (`GroupKFold` by administrative regency), coordinate-dependent models collapse completely ($F_1 \to 0.00$), whereas a Pure Biophysical Model excluding spatial coordinates preserves operational fire detection across unseen regencies (Holdout ROC-AUC: 0.7527; Extreme Fire Recall: 65.82%; Spatial CV mean $F_1 = 0.1399$, a 2.40-fold retention over coordinate models). 4. **Explainable AI and Environmental Tipping Points:** Exact Shapley value decomposition using TreeSHAP resolves critical non-linear tipping points: severe fire combustion escalates rapidly within 3,000 meters of artificial drainage canals and after 14 consecutive drought days. 5. **Actionable Societal Policy Reforms:** We translate remote sensing findings into concrete environmental governance: mandating statutory 3-km perimeter buffer stewardship for concessionaires operating contiguous peat canals, regularizing APL peatlands into community-managed Peatland Management Units (KPG) / Social Forestry, and prioritizing canal blocking along primary drainage corridors.

Editorial and Ethical Declarations:

- **Originality:** This manuscript is an original work that has not been published previously and is not under consideration for publication elsewhere.
- **Authorship and Affiliation:** The manuscript is co-authored by Ahmad Sopian (Independent Researcher, Samarinda, Kalimantan Timur, Indonesia; Corresponding Author) and Rahmi Susanti (Faculty of Public Health, Universitas Mulawarman, Samarinda, Kalimantan Timur, Indonesia).
- **Publishing Route:** We confirm submission under the standard **Subscription ($0 USD APC / Bebas Biaya Publikasi)** publishing route.
- **Funding Statement:** This research received no external funding or institutional grant and was conducted as an independent collaborative research project.
- **Conflict of Interest:** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Data Availability:** The master processed dataset of 43,937 active fire hotspots, official administrative boundaries (geoBoundaries ADM2), primary canal vectors in UTM Zone 49S projection, and complete analysis scripts are openly deposited and accessible in the research repository at https://github.com/amsopian22/karhutla-analisis.

Because this study speaks directly to the core mission of *Remote Sensing Applications: Society and Environment*—linking satellite remote sensing observations to critical environmental and societal governance solutions—we believe it will be of high interest to the journal's multidisciplinary readership.

Thank you very much for your time, consideration, and editorial handling of our manuscript.

Sincerely yours,

Ahmad Sopian (Corresponding Author)
Independent Researcher
Samarinda, Kalimantan Timur, Indonesia
E-mail: ahmadsopian22@gmail.com

Rahmi Susanti
Faculty of Public Health, Universitas Mulawarman
Samarinda, Kalimantan Timur, Indonesia
E-mail: rahmi.susanti@fkm.unmul.ac.id